

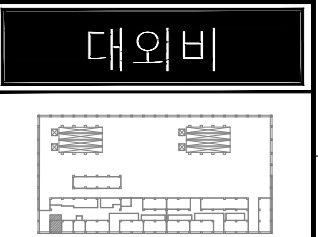
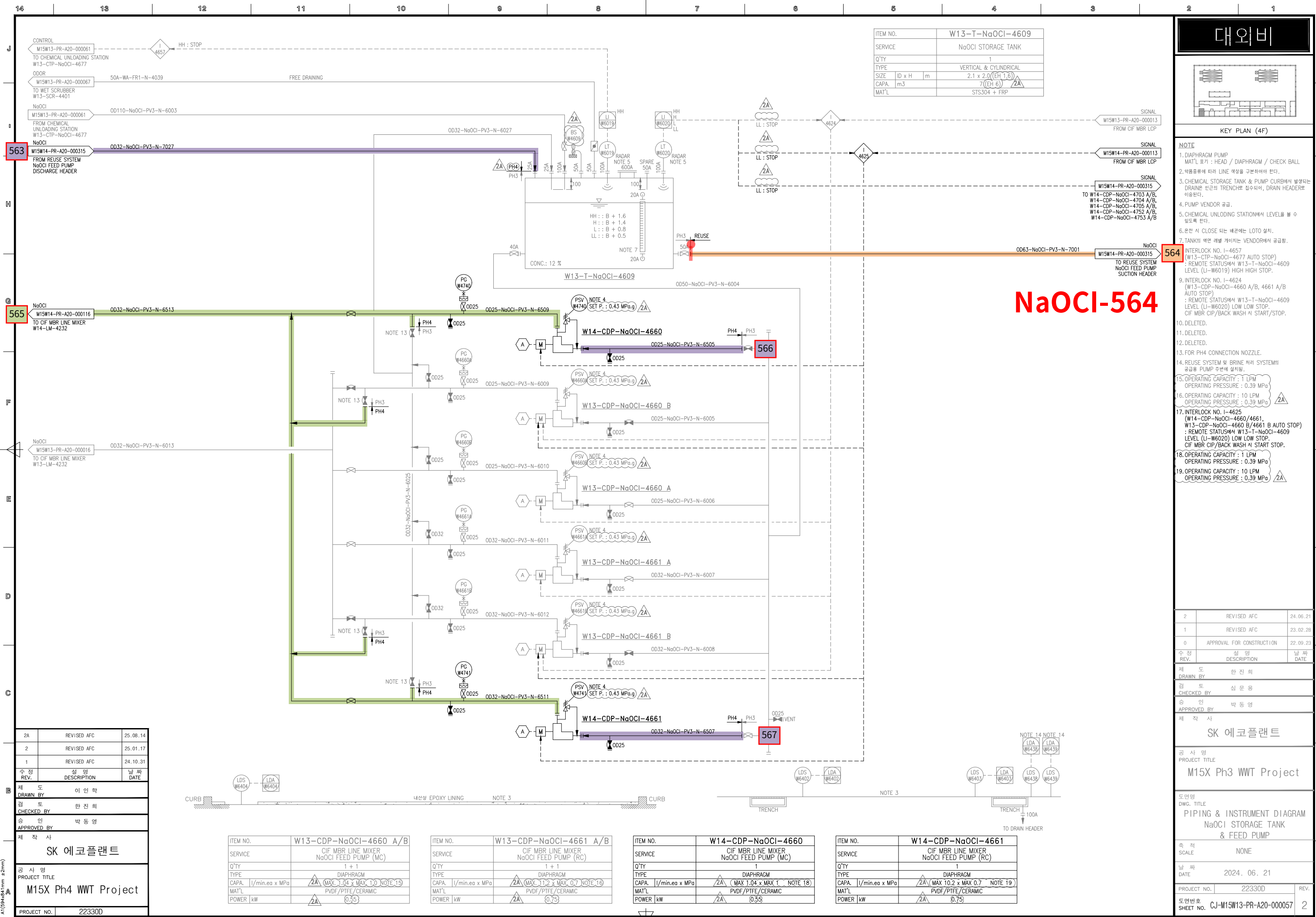
DRAWING LIST FOR TEST PACKAGE

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

NaOCI-564

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	564	4FA	OD25-NaOCI-PV3-N-7004-01	1 OF 1	0	CJ-M15W14-PR-A20-000315	1A	
2	564	4FA	OD25-NaOCI-PV3-N-7005-01	1 OF 1	0	CJ-M15W14-PR-A20-000315	1A	
3	564	4FA	OD25-NaOCI-PV3-N-7009-01	1 OF 1	0	CJ-M15W14-PR-A20-000315	1A	
4	564	4FA	OD25-NaOCI-PV3-N-7010-01	1 OF 1	0	CJ-M15W14-PR-A20-000315	1A	
5	564	4FA	OD32-NaOCI-PV3-N-7002-01	1 OF 1	0	CJ-M15W14-PR-A20-000315	1A	
6	564	4FA	OD32-NaOCI-PV3-N-7003-01	1 OF 1	0	CJ-M15W14-PR-A20-000315	1A	
7	564	4FA	OD32-NaOCI-PV3-N-7006-01	1 OF 1	0	CJ-M15W14-PR-A20-000315	1A	
8	564	4FA	OD32-NaOCI-PV3-N-7007-01	1 OF 1	0	CJ-M15W14-PR-A20-000315	1A	
9	564	4FA	OD32-NaOCI-PV3-N-7011-01	1 OF 1	0	CJ-M15W14-PR-A20-000315	1A	
10	564	4FA	OD32-NaOCI-PV3-N-7012-01	1 OF 1	0	CJ-M15W14-PR-A20-000315	1A	
11	564	4FA	OD63-NaOCI-PV3-N-7001-01	1 OF 1	0	CJ-M15W14-PR-A20-000315	1A	N 추가
12	564	4FA	OD63-NaOCI-PV3-N-7008-01	1 OF 1	0	CJ-M15W14-PR-A20-000315	1A	N 추가



- KEY PLAN (4F)
- NOTE
1. DIAPHRAGM PUMP
MAT'L 표기 : HEAD / DIAPHRAGM / CHECK BALL
 2. 약품종류에 따라 LINE 색상을 구분하여야 한다.
 3. CHEMICAL STORAGE TANK & PUMP CURB에서 발생하는 DRAIN에 인근의 TRENCH로 흡수되어, DRAIN HEADER로 이용된다.
 4. PUMP VENDOR 규격.
 5. CHEMICAL UNLOADING STATION에서 LEVEL을 볼 수 있도록 한다.
 6. 운전 시 CLOSE 되는 배관에는 LOTO 설치.
 7. TANK의 역면 레벨 게이지는 VENDOR에서 공급함.
 8. INTERLOCK NO. I-4657
(W13-CTP-NaOCI-4677 AUTO STOP)
: REMOTE STATUS에서 W13-T-NaOCI-4609 LEVEL (U-W6019) HIGH HIGH STOP.
 9. INTERLOCK NO. I-4624
(W13-CDP-NaOCI-4660 A/B, 4661 A/B AUTO STOP)
: REMOTE STATUS에서 W13-T-NaOCI-4609 LEVEL (U-W6020) LOW LOW STOP.
CIF MBR CIP/BACK WASH 시 START/STOP.
 10. DELETED.
 11. DELETED.
 12. DELETED.
 13. FOR PH4 CONNECTION NOZZLE.
 14. REUSE SYSTEM 및 BRINE 처리 SYSTEM의 공급용 PUMP 수단에 설치됨.
 15. OPERATING CAPACITY : 1 LPM
OPERATING PRESSURE : 0.39 MPa
 16. OPERATING CAPACITY : 10 LPM
OPERATING PRESSURE : 0.39 MPa
 17. INTERLOCK NO. I-4625
(W14-CDP-NaOCI-4660/4661, W13-CDP-NaOCI-4660 B/4661 B AUTO STOP)
: REMOTE STATUS에서 W13-T-NaOCI-4609 LEVEL (U-W6020) LOW LOW STOP.
CIF MBR CIP/BACK WASH 시 START STOP.
 18. OPERATING CAPACITY : 1 LPM
OPERATING PRESSURE : 0.39 MPa
 19. OPERATING CAPACITY : 10 LPM
OPERATING PRESSURE : 0.39 MPa

2A	REVISED AFC	25.08.14
2	REVISED AFC	25.01.17
1	REVISED AFC	24.10.31
수정 REV.	설명 DESCRIPTION	날짜 DATE
제 도	이 인 학	
DRAWN BY		
검 토	한 진 희	
CHECKED BY		
승 인	박 동 영	
APPROVED BY		
제 작 사	SK 에코플랜트	
공 사 명	M15X Ph4 WWT Project	
PROJECT TITLE		
PROJECT NO.	22330D	

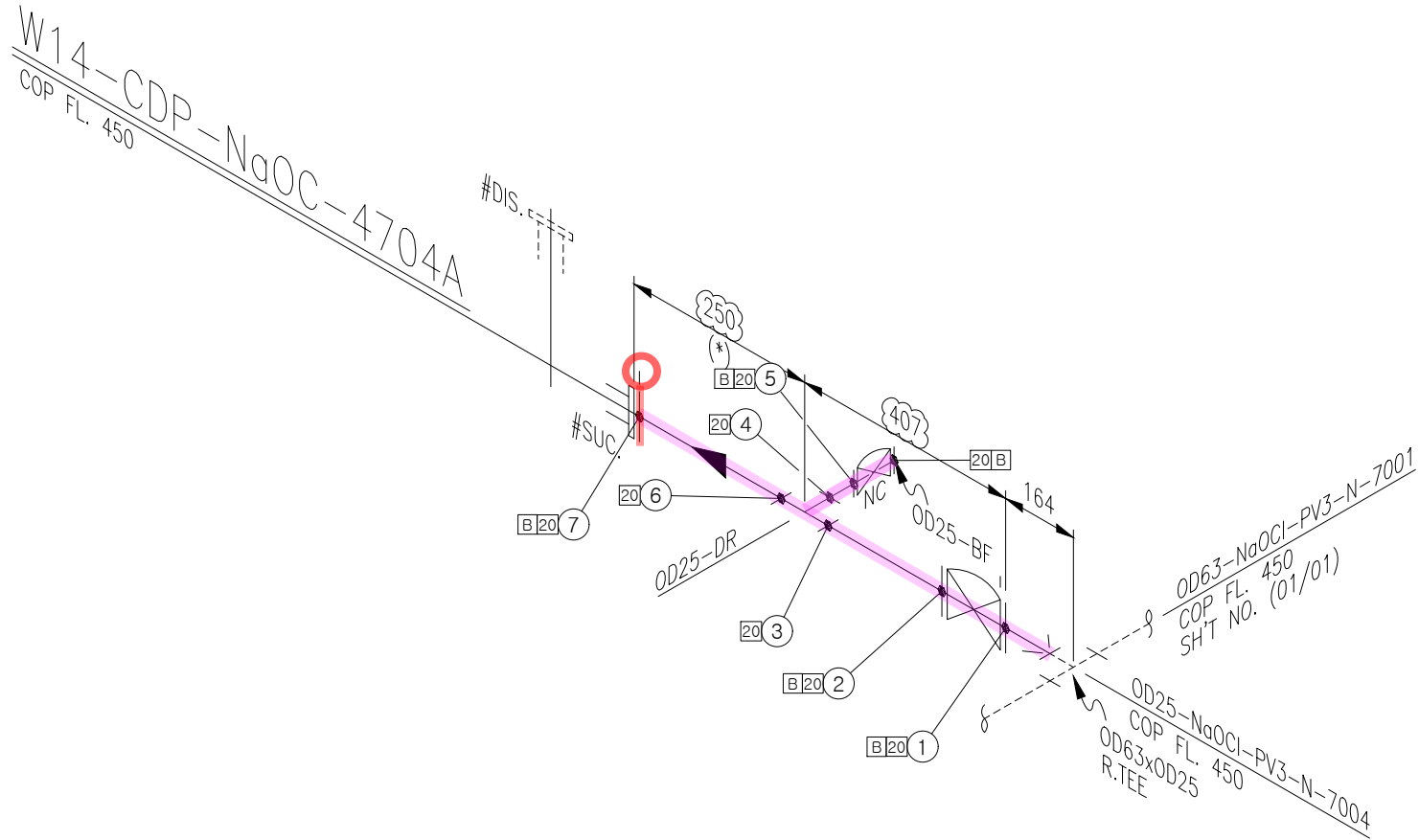
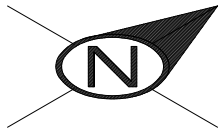
ITEM NO.	W13-CDP-NaOCI-4660 A/B
SERVICE	CIF MBR LINE MIXER NaOCI FEED PUMP (MC)
Q'TY	1 + 1
TYPE	DIAPHRAGM
CAPA.	1/min.ea x MPa 2A (MAX 1.04 x MAX 1.0 NOTE 19)
MAT'L	PVDF/PTFE/CERAMIC 2A (0.55)
POWER kW	2A (0.55)

ITEM NO.	W13-CDP-NaOCI-4661 A/B
SERVICE	CIF MBR LINE MIXER NaOCI FEED PUMP (RC)
Q'TY	1 + 1
TYPE	DIAPHRAGM
CAPA.	1/min.ea x MPa 2A (MAX 1.04 x MAX 0.7 NOTE 19)
MAT'L	PVDF/PTFE/CERAMIC 2A (0.75)
POWER kW	2A (0.75)

ITEM NO.	W14-CDP-NaOCI-4660
SERVICE	CIF MBR LINE MIXER NaOCI FEED PUMP (MC)
Q'TY	1
TYPE	DIAPHRAGM
CAPA.	1/min.ea x MPa 2A (MAX 1.04 x MAX 1.0 NOTE 19)
MAT'L	PVDF/PTFE/CERAMIC 2A (0.55)
POWER kW	2A (0.55)

ITEM NO.	W14-CDP-NaOCI-4661
SERVICE	CIF MBR LINE MIXER NaOCI FEED PUMP (RC)
Q'TY	1
TYPE	DIAPHRAGM
CAPA.	1/min.ea x MPa 2A (MAX 10.2 x MAX 0.7 NOTE 19)
MAT'L	PVDF/PTFE/CERAMIC 2A (0.75)
POWER kW	2A (0.75)

2	REVISED AFC	24.06.21
1	REVISED AFC	23.02.28
0	APPROVAL FOR CONSTRUCTION	22.09.23
수정 REV.	설명 DESCRIPTION	날짜 DATE
제 도	한 진 희	
DRAWN BY		
검 토	심 문 용	
CHECKED BY		
승 인	박 동 영	
APPROVED BY		
제 작 사	SK 에코플랜트	
공 사 명	M15X Ph3 WWT Project	
PROJECT TITLE		
도면명 DWG. TITLE	PIPING & INSTRUMENT DIAGRAM NaOCI STORAGE TANK & FEED PUMP	
속 척 SCALE	NONE	
날 짜 DATE	2024. 06. 21	
PROJECT NO.	22330D	REV.
도면번호 SHEET NO.	CJ-M15W13-PR-A20-000057	2



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<NOTE>
1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLQ/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1



ITEM	SIZE	Q'TY	DESCRIPTION
PV3			
P	OD25	0.8M	PIPE, CPVC, PE, SOCKET FUSION, SDR13.5
T	OD25	1EA	TEE EQUAL, CPVC, PE, SOCKET FUSION, SDR13.5
F	OD25	4EA	FLG, KS 10K, FF, STUB FLG CPVC /
			BACKING RING(PP)+INSERT(STEEL), SDR13.5
FB	OD25	1EA	FLG BLIND, KS 10K, FF, CPVC, SDR13.5
G	OD25	5EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
B	OD25	5SET	MACHINE BOLT & NYLON NUT :STS 304 / W WASHERS
VD	OD25	2EA	DIAPHRAGM VALVE, KS 10K, FF FLG'D, BODY-CPVC /
SH	OD25	5EA	SHIELD
LOTO	OD25	1EA	LOTO

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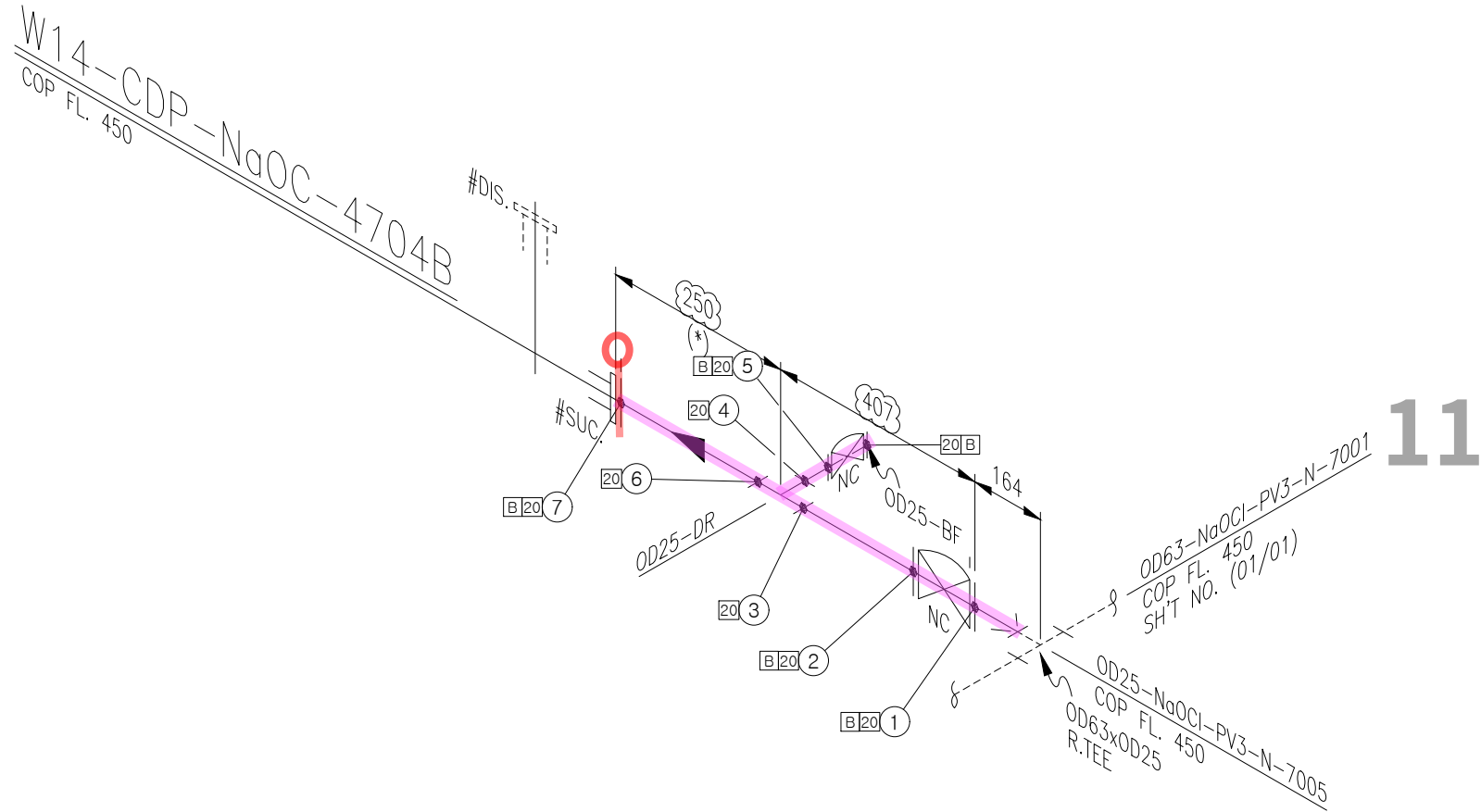


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CONTRACTOR		PROJECT NAME	
SK ecoplant		M15X Ph4 Project	
COMMODITY		JOB NO.	
NaOCI		22330D	
LINE NO.		SH'T NO.	REV.
OD25-NaOCI-PV3-N-7004		01 / 01	0



ITEM	SIZE	Q'TY	DESCRIPTION
		PV3	
P	OD25	0.8M	PIPE, CPVC, PE, SOCKET FUSION, SDR13.5
T	OD25	1EA	TEE EQUAL, CPVC, PE, SOCKET FUSION, SDR13.5
F	OD25	4EA	FLG, KS 10K, FF, STUB FLG CPVC / BACKING RING(PP)+INSERT(STEEL), SDR13.5
FB	OD25	1EA	FLG BLIND, KS 10K, FF, CPVC, SDR13.5
G	OD25	5EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
B	OD25	5SET	MACHINE BOLT & NYLON NUT :STS 304 / W WASHERS
VD	OD25	2EA	DIAPHRAGM VALVE, KS 10K, FF FLG'D, BODY-CPVC /
SH	OD25	5EA	SHIELD
LOTO	OD25	2EA	LOTO



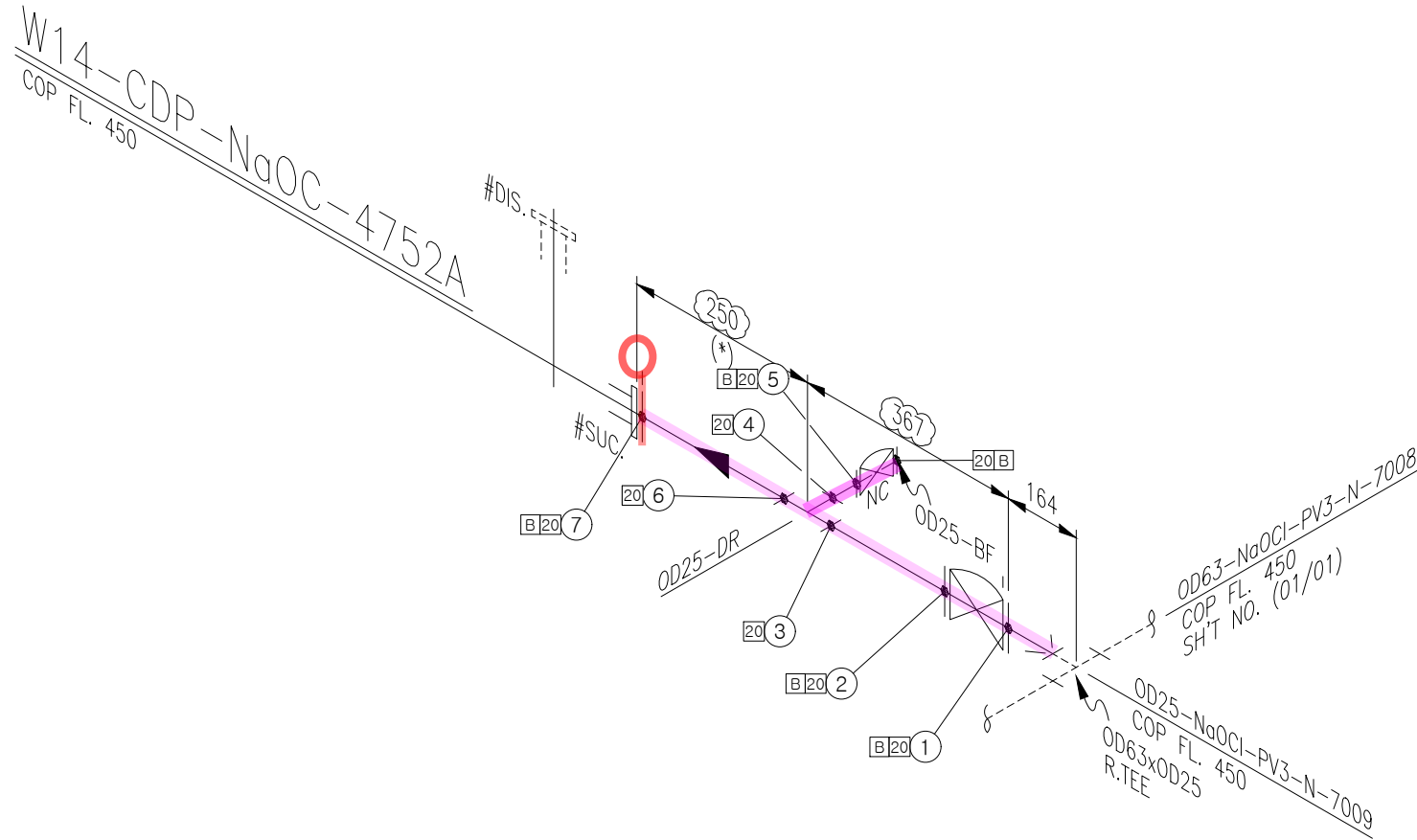
<NOTE>
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB1/VB10/VB2/VB210	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VD10	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

[illegible]



ITEM	SIZE	Q'TY	DESCRIPTION
		PV3	
P	OD25	0.8M	PIPE, CPVC, PE, SOCKET FUSION, SDR13.5
T	OD25	1EA	TEE EQUAL, CPVC, PE, SOCKET FUSION, SDR13.5
F	OD25	4EA	FLG, KS 10K, FF, STUB FLG CPVC /
			BACKING RING(PP)+INSERT(STEEL), SDR13.5
FB	OD25	1EA	FLG BLIND, KS 10K, FF, CPVC, SDR13.5
G	OD25	5EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
B	OD25	5SET	MACHINE BOLT & NYLON NUT :STS 304 / W WASHERS
VD	OD25	2EA	DIAPHRAGM VALVE, KS 10K, FF FLG'D, BODY-CPVC /
SH	OD25	5EA	SHIELD
LOTO	OD25	1EA	LOTO



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<NOTE>

1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB1/VB2/VB2L0	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDL0	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

CLIENT

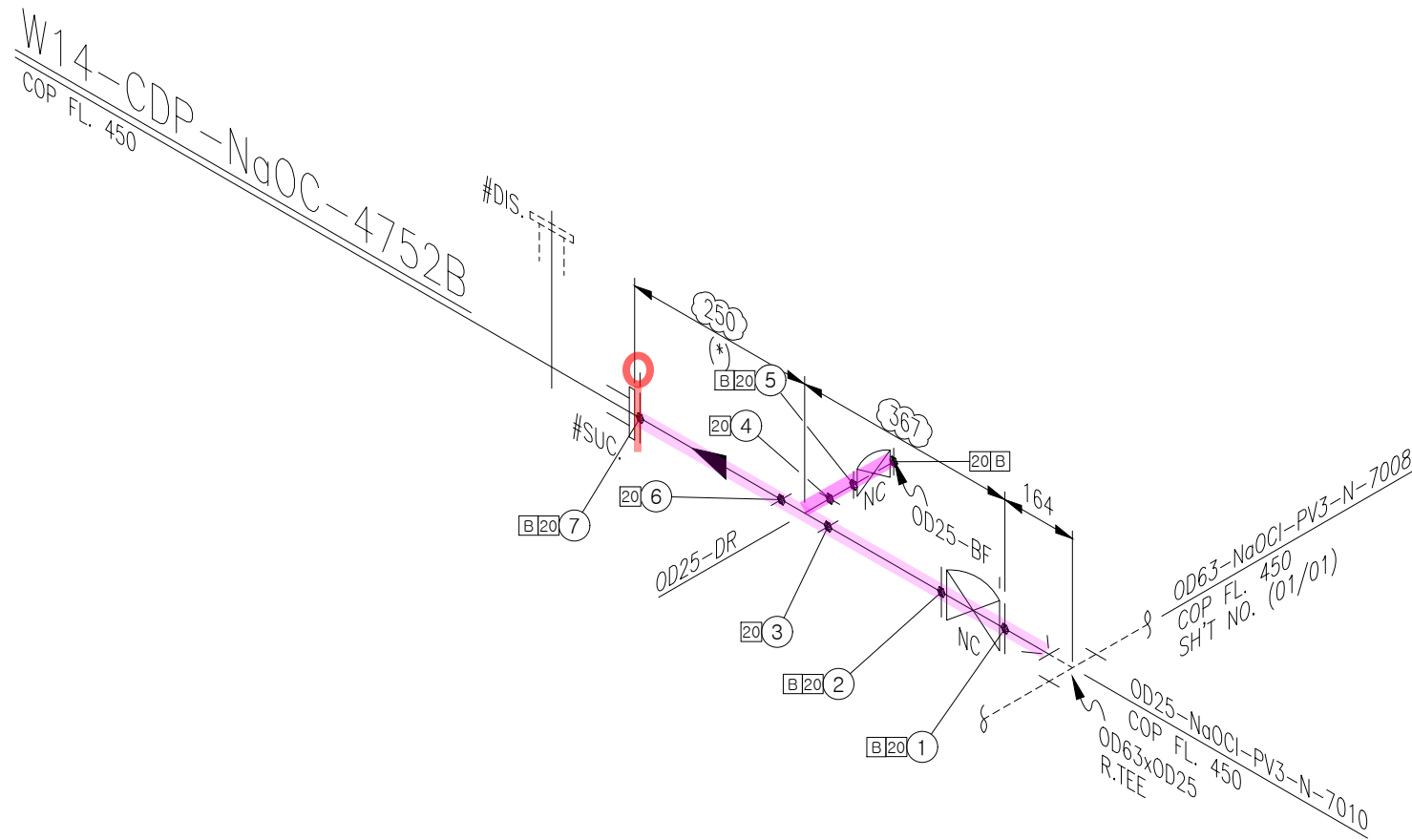


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INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	1.0	60	UNIT 4FA								M15X Ph4 Project	
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040010-A		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY	NaOCl	JOB NO.
						REF. P & ID CJ-M15W14-PR-A20-000315									
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D			



ITEM	SIZE	Q'TY	DESCRIPTION
		PV3	
P	OD25	0.8M	PIPE, CPVC, PE, SOCKET FUSION, SDR13.5
T	OD25	1EA	TEE EQUAL, CPVC, PE, SOCKET FUSION, SDR13.5
F	OD25	4EA	FLG, KS 10K, FF, STUB FLG CPVC /
			BACKING RING(PP)+INSERT(STEEL), SDR13.5
FB	OD25	1EA	FLG BLIND, KS 10K, FF, CPVC, SDR13.5
G	OD25	5EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
B	OD25	5SET	MACHINE BOLT & NYLON NUT :STS 304 / W WASHERS
VD	OD25	2EA	DIAPHRAGM VALVE, KS 10K, FF FLG'D, BODY-CPVC /
SH	OD25	5EA	SHIELD
LOTO	OD25	2EA	LOTO



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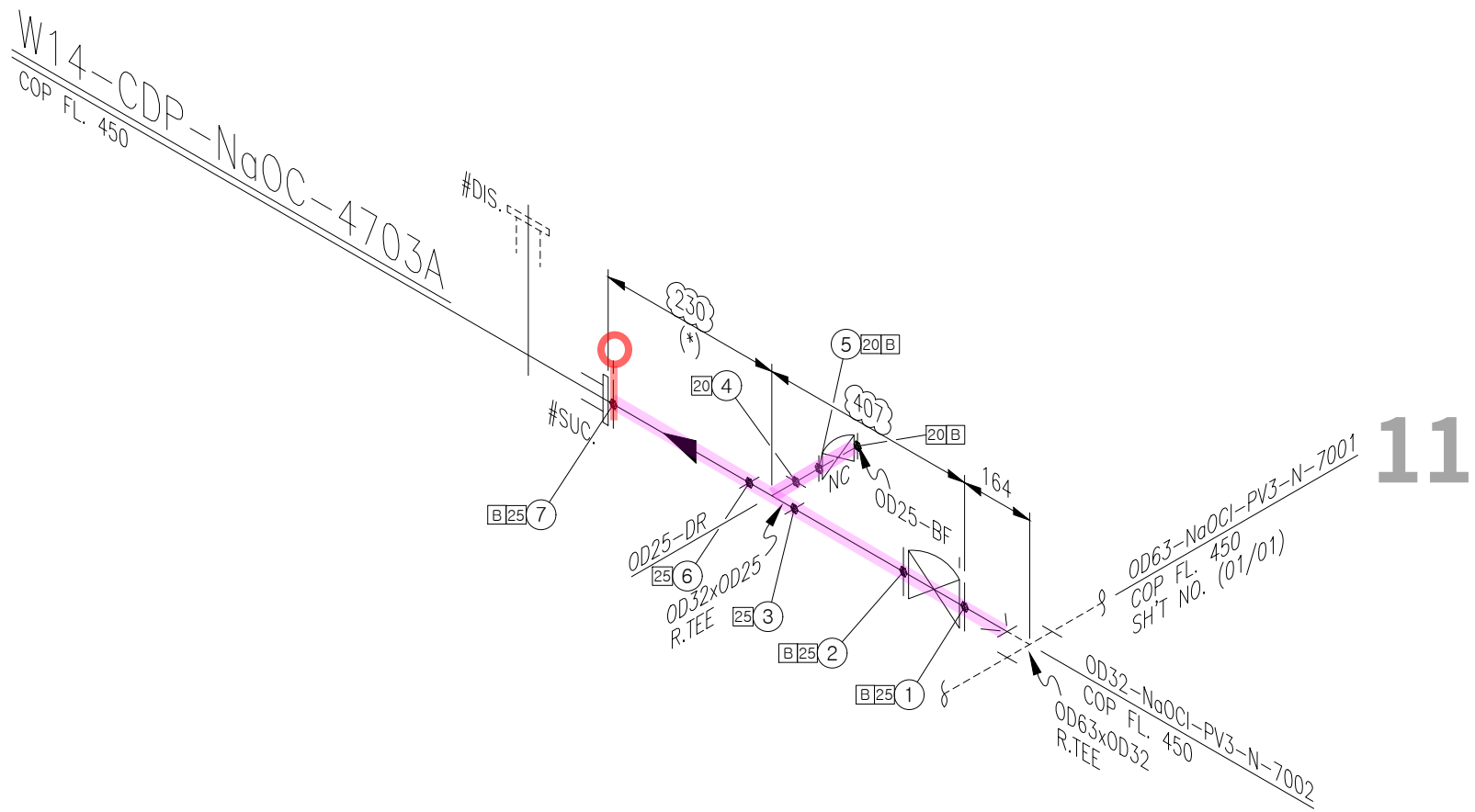
- <NOTE>
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm ² g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	1.0	60	UNIT	4FA								M15X Ph4 Project	
			OPERATING	ATM	AMB	REF. PLAN DWG.								COMMODITY	JOB NO.	
						CJ-M15W14-PI-A30-040010-A			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	NaOCl	22330D	
						REF. P & ID		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO.	REV.
						CJ-M15W14-PR-A20-000315								0D25-NaOCl-PV3-N-7010	01 / 01	0



ITEM	SIZE	Q'TY	DESCRIPTION
		0.7M	
P	OD32	0.7M	PIPE, CPVC, PE, SOCKET FUSION, SDR13.5
P	OD25	0.1M	PIPE, CPVC, PE, SOCKET FUSION, SDR13.5
TR	OD32xOD25	1EA	TEE RED, CPVC, PE, SOCKET FUSION, SDR13.5
F	OD32	3EA	FLG, KS 10K, FF, STUB FLG CPVC / BACKING RING(PP)+INSERT(STEEL), SDR13.5
F	OD25	1EA	FLG, KS 10K, FF, STUB FLG CPVC / BACKING RING(PP)+INSERT(STEEL), SDR13.5
FB	OD25	1EA	FLG BLIND, KS 10K, FF, CPVC, SDR13.5
G	OD32	3EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G	OD25	2EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
B	OD32	3SET	MACHINE BOLT & NYLON NUT :STS 304 / W WASHERS
B	OD25	2SET	MACHINE BOLT & NYLON NUT :STS 304 / W WASHERS
VD	OD32	1EA	DIAPHRAGM VALVE, KS 10K, FF FLG'D, BODY-CPVC / TRIM-PDDM, BB, HW
VD	OD25	1EA	DIAPHRAGM VALVE, KS 10K, FF FLG'D, BODY-CPVC / TRIM-PDDM, BB, HW
SH	OD32	3EA	SHIELD
SH	OD25	2EA	SHIELD
LOTO	OD25	1EA	LOTO



<NOTE>
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

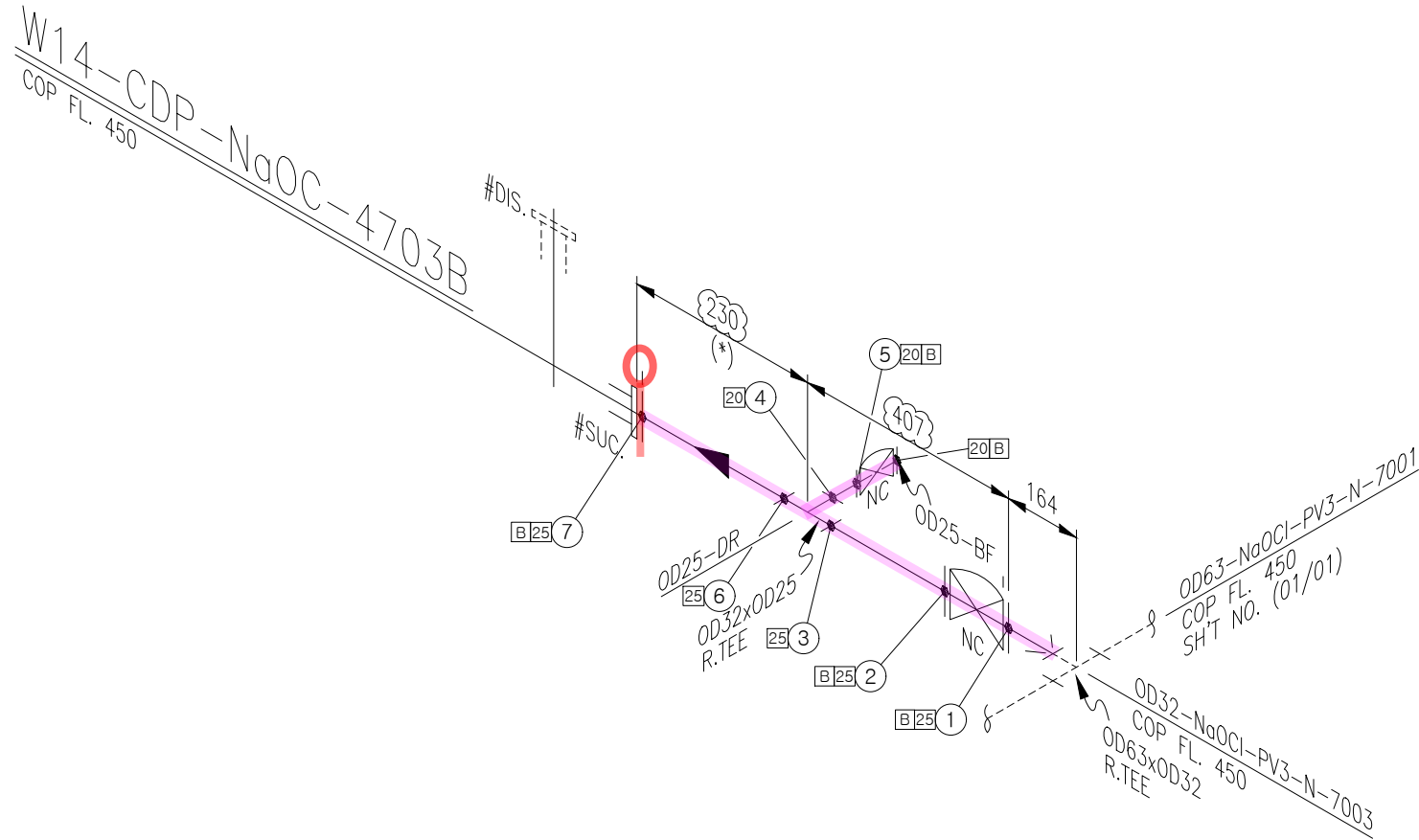
CLIENT				5	
CONTRACTOR			PROJECT NAME		
			M15X Ph4 Project		
COMMODITY			JOB NO.		
NaOCI			22330D		
LINE NO.			SH'T NO.		REV.
0D32-NaOCI-PV3-N-7002			01 / 01		0

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB0/VB1/VB1L0/VB2/VB2L0	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VCS/VCS/VCS					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDL0	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME			
	THICK.	—	DESIGN	1.0	60	UNIT	4FA								M15X Ph4 Project			
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040010-A								COMMODITY	NaOCl			
						REF. P & ID CJ-M15W14-PR-A20-000315			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	0D32-NaOCl-PV3-N-7002		SH'T NO. 01 / 01	REV. 0
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D					



ITEM	SIZE	Q'TY	DESCRIPTION
		PV3	
P	OD32	0.7M	PIPE, CPVC, PE, SOCKET FUSION, SDR13.5
P	OD25	0.1M	PIPE, CPVC, PE, SOCKET FUSION, SDR13.5
TR	OD32xOD25	1EA	TEE RED, CPVC, PE, SOCKET FUSION, SDR13.5
F	OD32	3EA	FLG, KS 10K, FF, STUB FLG CPVC /
			BACKING RING(PP)+INSERT(STEEL), SDR13.5
F	OD25	1EA	FLG, KS 10K, FF, STUB FLG CPVC /
			BACKING RING(PP)+INSERT(STEEL), SDR13.5
FB	OD25	1EA	FLG BLIND, KS 10K, FF, CPVC, SDR13.5
G	OD32	3EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G	OD25	2EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
B	OD32	3SET	MACHINE BOLT & NYLON NUT :STS 304 / W WASHERS
B	OD25	2SET	MACHINE BOLT & NYLON NUT :STS 304 / W WASHERS
VD	OD32	1EA	DIAPHRAGM VALVE, KS 10K, FF FLG'D, BODY-CPVC /
			TRIM-PDDM, BB, HW
VD	OD25	1EA	DIAPHRAGM VALVE, KS 10K, FF FLG'D, BODY-CPVC /
			TRIM-PDDM, BB, HW
SH	OD32	3EA	SHIELD
SH	OD25	2EA	SHIELD
LOTO	OD25	1EA	LOTO
LOTO	OD32	1EA	LOTO



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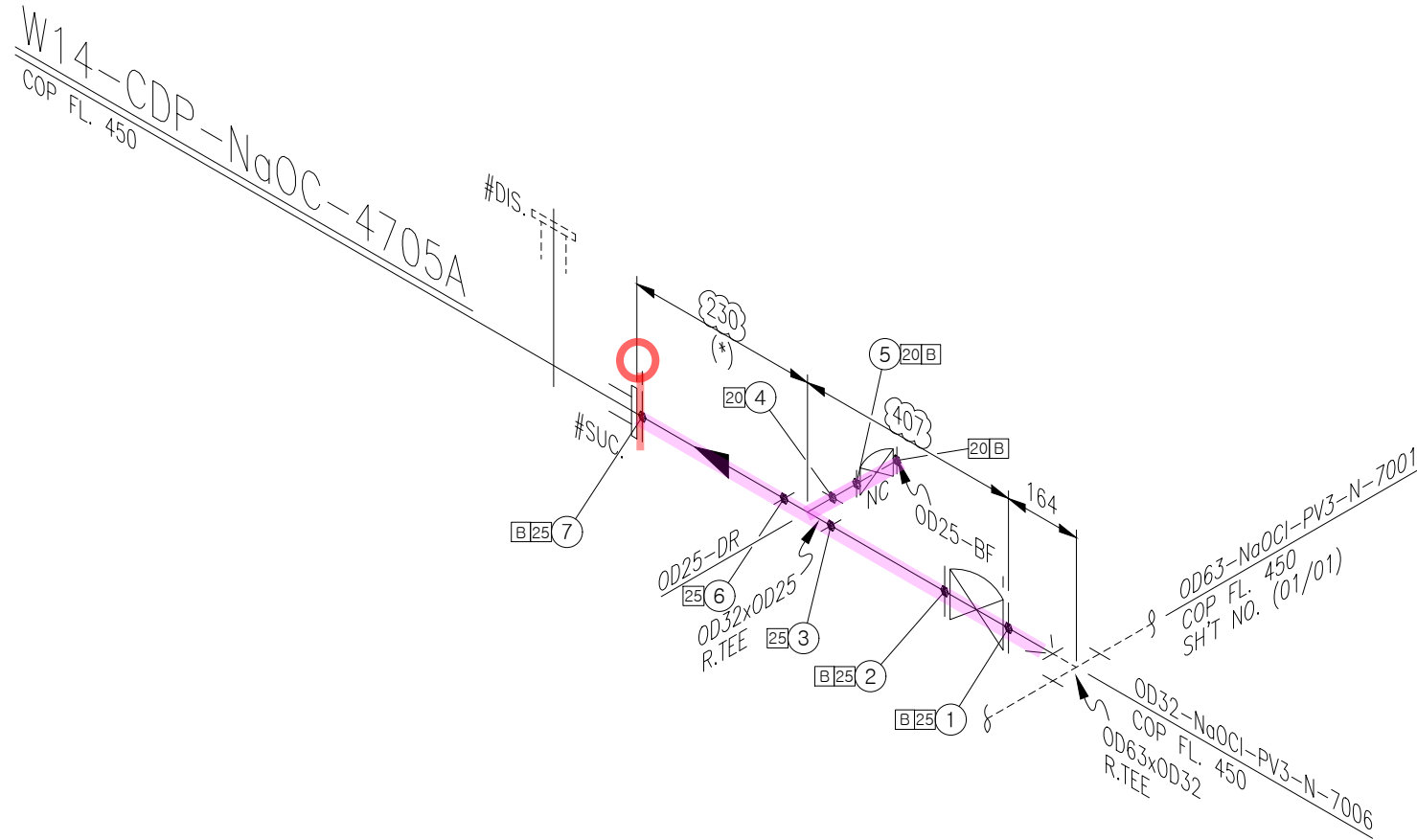
- <NOTE>
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	1.0	60	UNIT 4FA								M15X Ph4 Project	
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040010-A		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY NaOCI	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000315							REV. NO.	DATE	DESCRIPTION



ITEM	SIZE	Q'TY	DESCRIPTION
		PV3	
P	OD32	0.8M	PIPE, CPVC, PE, SOCKET FUSION, SDR13.5
P	OD25	0.1M	PIPE, CPVC, PE, SOCKET FUSION, SDR13.5
TR	OD32xOD25	1EA	TEE RED, CPVC, PE, SOCKET FUSION, SDR13.5
F	OD32	3EA	FLG, KS 10K, FF, STUB FLG CPVC / BACKING RING(PP)+INSERT(STEEL), SDR13.5
F	OD25	1EA	FLG, KS 10K, FF, STUB FLG CPVC / BACKING RING(PP)+INSERT(STEEL), SDR13.5
FB	OD25	1EA	FLG BLIND, KS 10K, FF, CPVC, SDR13.5
G	OD32	3EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G	OD25	2EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
B	OD32	3SET	MACHINE BOLT & NYLON NUT :STS 304 / W WASHERS
B	OD25	2SET	MACHINE BOLT & NYLON NUT :STS 304 / W WASHERS
VD	OD32	1EA	DIAPHRAGM VALVE, KS 10K, FF FLG'D, BODY-CPVC / TRIM-PDDM, BB, HW
VD	OD25	1EA	DIAPHRAGM VALVE, KS 10K, FF FLG'D, BODY-CPVC / TRIM-PDDM, BB, HW
SH	OD32	3EA	SHIELD
SH	OD25	2EA	SHIELD
LOTO	OD25	1EA	LOTO



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<NOTE>

1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

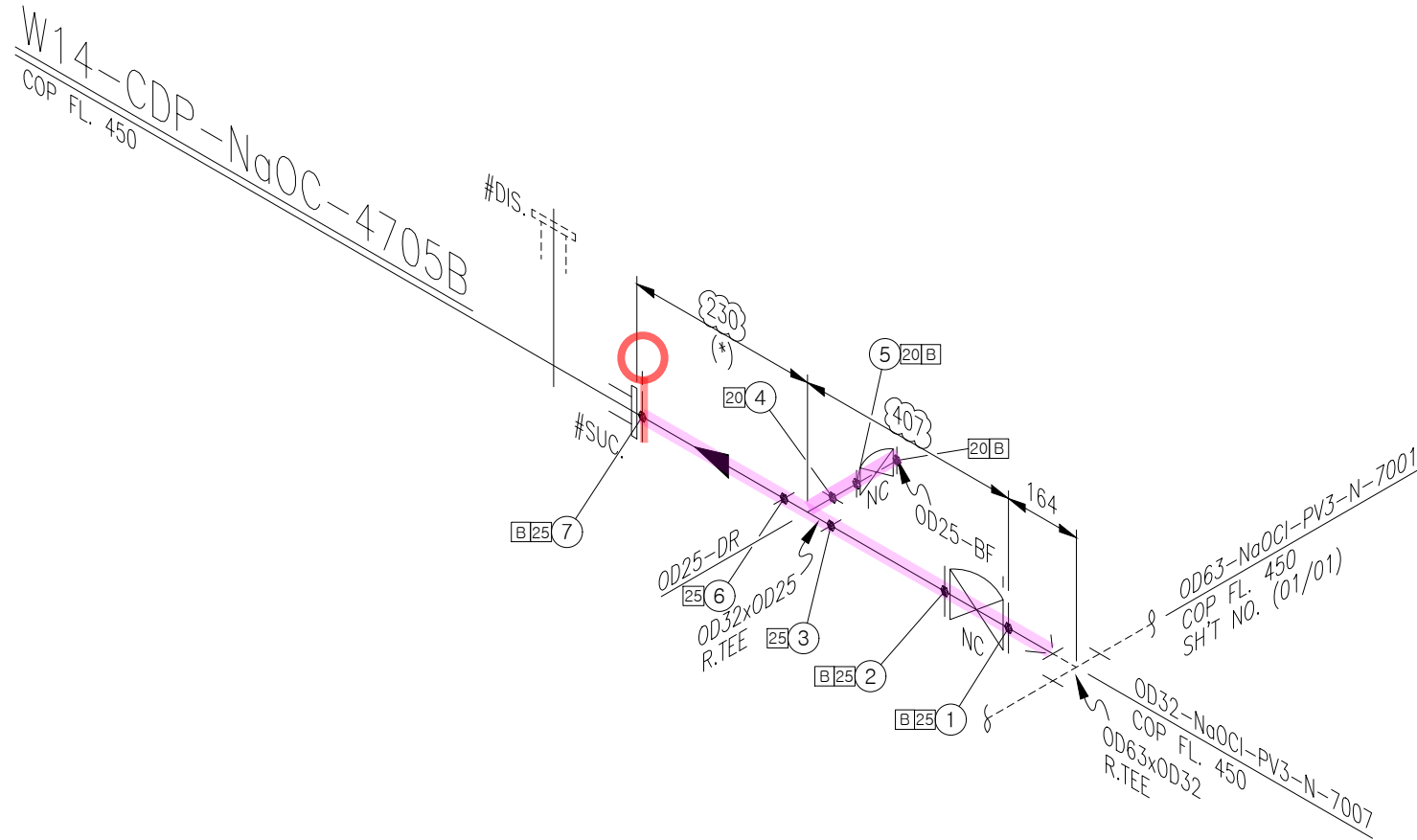
* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14									CONTRACTOR	PROJECT NAME		
	THICK.	—	DESIGN	1.0	60	UNIT	4FA										M15X Ph4 Project		
			OPERATING	ATM	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-040010-A		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영		COMMODITY	NaOCl	JOB NO.		
						REF. P & ID	CJ-M15W14-PR-A20-000315								REV. NO.	DATE	DESCRIPTION	BY	CHK
																	22330D	01 / 01	0

경고 : 본 문건은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 무단 복제·배포, 제3자에게 공개, 누설 또는 기타 방법으로 유출하여서는 안되며, 회사는 은포자의 귀책사유로 인해 피해가 발생할 경우 은포자에게 민/형사상 책임을 물을 수 있습니다.



ITEM	SIZE	Q'TY	DESCRIPTION
PV3			
P	OD32	0.8M	PIPE, CPVC, PE, SOCKET FUSION, SDR13.5
P	OD25	0.1M	PIPE, CPVC, PE, SOCKET FUSION, SDR13.5
TR	OD32xOD25	1EA	TEE RED, CPVC, PE, SOCKET FUSION, SDR13.5
F	OD32	3EA	FLG, KS 10K, FF, STUB FLG CPVC /
			BACKING RING(PP)+INSERT(STEEL), SDR13.5
F	OD25	1EA	FLG, KS 10K, FF, STUB FLG CPVC /
			BACKING RING(PP)+INSERT(STEEL), SDR13.5
FB	OD25	1EA	FLG BLIND, KS 10K, FF, CPVC, SDR13.5
G	OD32	3EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G	OD25	2EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
B	OD32	3SET	MACHINE BOLT & NYLON NUT :STS 304 / W WASHERS
B	OD25	2SET	MACHINE BOLT & NYLON NUT :STS 304 / W WASHERS
VD	OD32	1EA	DIAPHRAGM VALVE, KS 10K, FF FLG'D, BODY-CPVC /
			TRIM-PDDM, BB, HW
VD	OD25	1EA	DIAPHRAGM VALVE, KS 10K, FF FLG'D, BODY-CPVC /
			TRIM-PDDM, BB, HW
SH	OD32	3EA	SHIELD
SH	OD25	2EA	SHIELD
LOTO	OD25	1EA	LOTO
LOTO	OD32	1EA	LOTO



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<NOTE>

1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

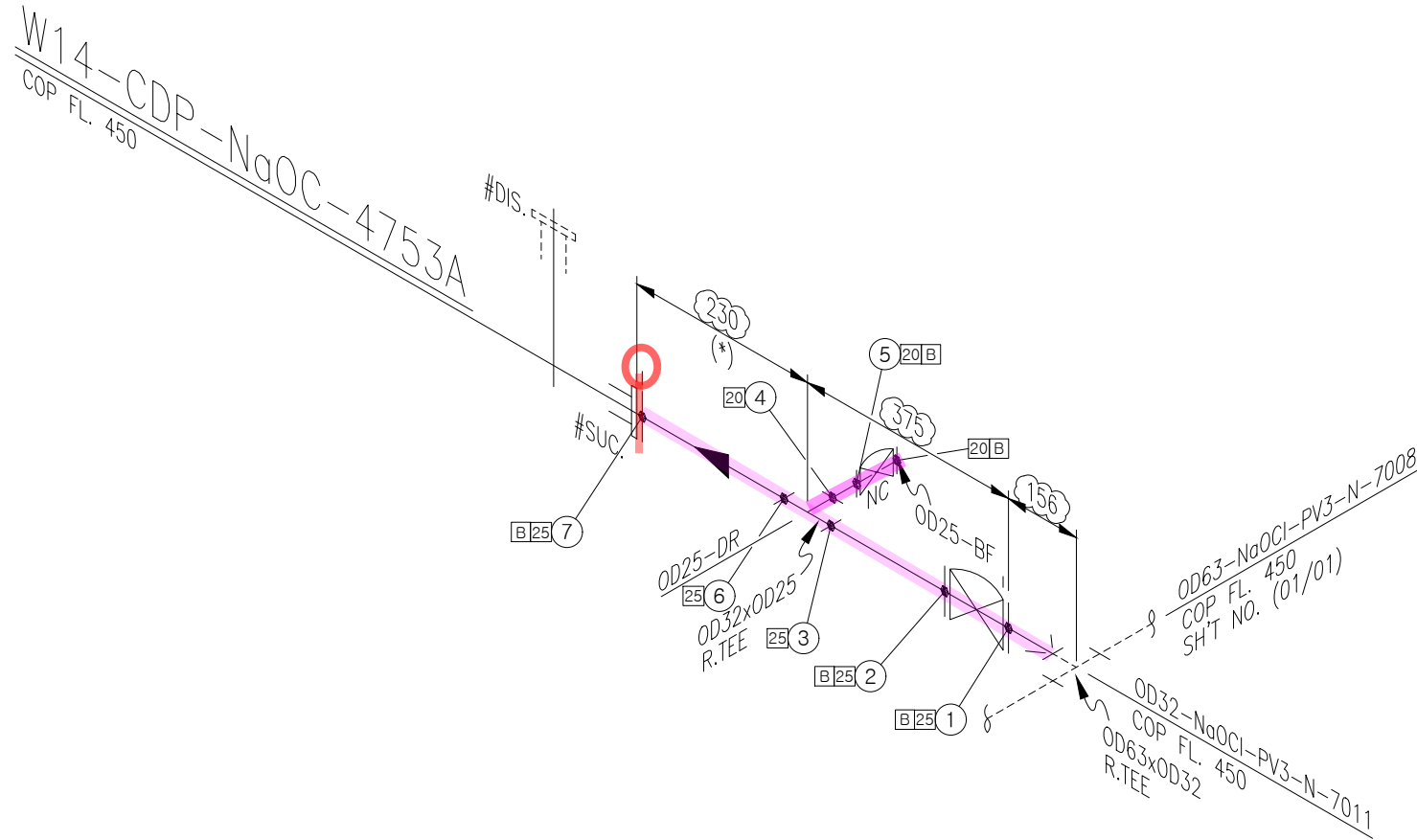
* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

CLIENT  8

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm ² g)	TEMP. (°C)	AREA W14							CONTRACTOR 	PROJECT NAME	
	THICK.	-	DESIGN	1.0	60	UNIT 4FA								M15X Ph4 Project	
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040010-A							COMMODITY NaOCl	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000315		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO. 0D32-No0Cl-PV3-N-7007	SH'T NO. 01 / 01	REV. 0
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D			



ITEM	SIZE	Q'TY	DESCRIPTION
PV3			
P	OD32	0.8M	PIPE, CPVC, PE, SOCKET FUSION, SDR13.5
P	OD25	0.1M	PIPE, CPVC, PE, SOCKET FUSION, SDR13.5
TR	OD32xOD25	1EA	TEE RED, CPVC, PE, SOCKET FUSION, SDR13.5
F	OD32	3EA	FLG, KS 10K, FF, STUB FLG CPVC /
			BACKING RING(PP)+INSERT(STEEL), SDR13.5
F	OD25	1EA	FLG, KS 10K, FF, STUB FLG CPVC /
			BACKING RING(PP)+INSERT(STEEL), SDR13.5
FB	OD25	1EA	FLG BLIND, KS 10K, FF, CPVC, SDR13.5
G	OD32	3EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G	OD25	2EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
B	OD32	3SET	MACHINE BOLT & NYLON NUT :STS 304 / W WASHERS
B	OD25	2SET	MACHINE BOLT & NYLON NUT :STS 304 / W WASHERS
VD	OD32	1EA	DIAPHRAGM VALVE, KS 10K, FF FLG'D, BODY-CPVC /
			TRIM-PDDM, BB, HW
VD	OD25	1EA	DIAPHRAGM VALVE, KS 10K, FF FLG'D, BODY-CPVC /
			TRIM-PDDM, BB, HW
SH	OD32	3EA	SHIELD
SH	OD25	2EA	SHIELD
LOTO	OD25	1EA	LOTO



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<NOTE>

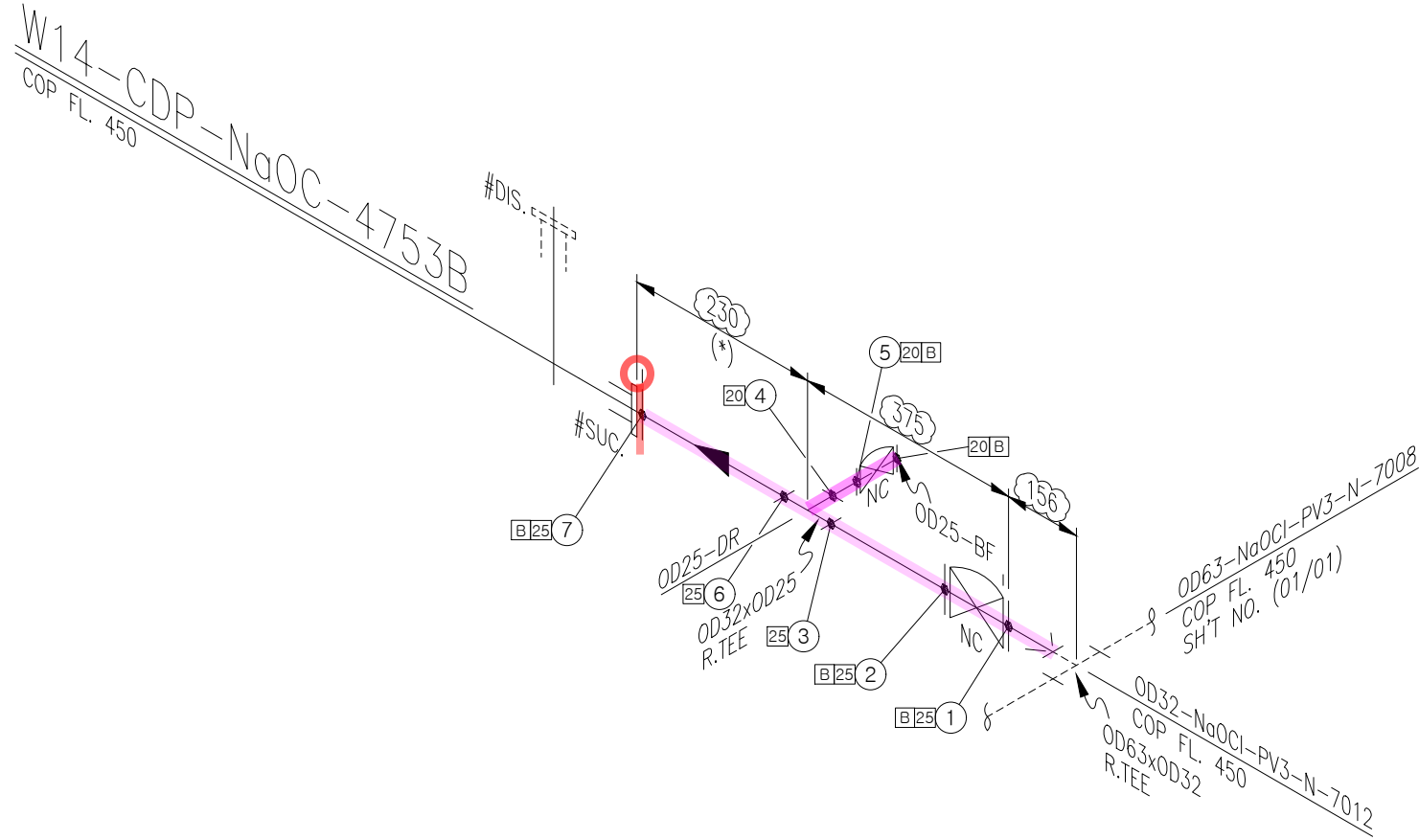
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB1/VB2/VB2L0	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDL0	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14								CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	1.0	60	UNIT 4FA									M15X Ph4 Project	
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040010-A		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY	NaOCI	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000315							REV. NO.	DATE	DESCRIPTION	BY



ITEM	SIZE	Q'TY	DESCRIPTION
PV3			
P	OD32	0.8M	PIPE, CPVC, PE, SOCKET FUSION, SDR13.5
P	OD25	0.1M	PIPE, CPVC, PE, SOCKET FUSION, SDR13.5
TR	OD32xOD25	1EA	TEE RED, CPVC, PE, SOCKET FUSION, SDR13.5
F	OD32	3EA	FLG, KS 10K, FF, STUB FLG CPVC /
			BACKING RING(PP)+INSERT(STEEL), SDR13.5
F	OD25	1EA	FLG, KS 10K, FF, STUB FLG CPVC /
			BACKING RING(PP)+INSERT(STEEL), SDR13.5
FB	OD25	1EA	FLG BLIND, KS 10K, FF, CPVC, SDR13.5
G	OD32	3EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G	OD25	2EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
B	OD32	3SET	MACHINE BOLT & NYLON NUT :STS 304 / W WASHERS
B	OD25	2SET	MACHINE BOLT & NYLON NUT :STS 304 / W WASHERS
VD	OD32	1EA	DIAPHRAGM VALVE, KS 10K, FF FLG'D, BODY-CPVC /
			TRIM-PDDM, BB, HW
VD	OD25	1EA	DIAPHRAGM VALVE, KS 10K, FF FLG'D, BODY-CPVC /
			TRIM-PDDM, BB, HW
SH	OD32	3EA	SHIELD
SH	OD25	2EA	SHIELD
LOTO	OD25	1EA	LOTO
LOTO	OD32	1EA	LOTO



12

<NOTE>
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

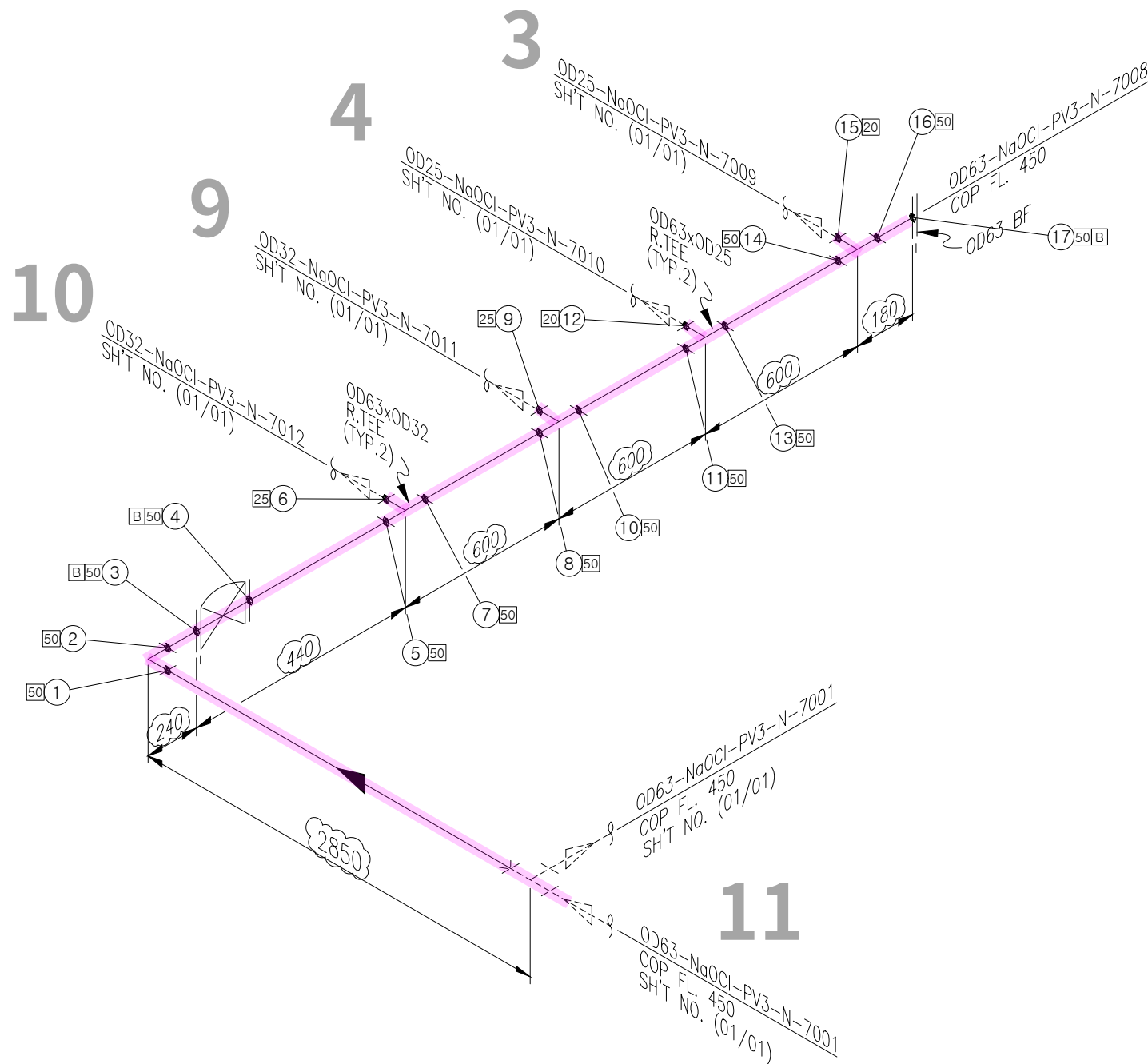
* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	1.0	60	UNIT 4FA								M15X Ph4 Project	
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040010-A		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY NaOCI	JOB NO. 22330D	
					REF. P & ID CJ-M15W14-PR-A20-000315	LINE NO. OD32-NaOCI-PV3-N-7012							SH'T NO. 01 / 01	REV. 0	
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D			

조각 : SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 제3자에게 인계, 개제, 제3자에게 양도할 수 없으며, 회사나 유무포자의 사명상 이익을 위하여 유무포자에게 인/항사자책할 수 있는



ITEM	SIZE	Q'TY	DESCRIPTION
PV3			
P	OD63	5.5M	PIPE, CPVC, PE, SOCKET FUSION, SDR13.5
E90	OD63	1EA	ELBOW 90 DEG, CPVC, PE, SOCKET FUSION, SDR13.5
TR	OD63xOD32	2EA	TEE RED, CPVC, PE, SOCKET FUSION, SDR13.5
TR	OD63xOD25	2EA	TEE RED, CPVC, PE, SOCKET FUSION, SDR13.5
F	OD63	3EA	FLG, KS 10K, FF, STUB FLG CPVC /
FB	OD63	1EA	FLG BLIND RING (FR) INSERT (STEEL) SDR13.5
G	OD63	3EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
B	OD63	3SET	MACHINE BOLT & NYLON NUT :STS 304 / W WASHERS
VD	OD63	1EA	DIAPHRAGM VALVE, KS 10K, FF FLG'D, BODY-CPVC /
			TRIM-PDDM, BB, HW
SH	OD63	3EA	SHIELD



<NOTE>

1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB1/VB10/VB2/VB210	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VD10	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR 		PROJECT NAME		
	THICK.	—	DESIGN	1.0	60	UNIT 4FA							COMMODITY NaOCl		M15X Ph4 Project		
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040009-A							JOB NO.		22330D		
								25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO. 0D63-NaOCl-PV3-N-7008		SH'T NO. 01 / 01		REV. 0
						REF. P & ID CJ-M15W14-PR-A20-000315	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D					

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누출 및 유출될 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음